

1. (5 pts)

a) What is the dimension of a vector space?

b) What is the rank of a matrix A ?

2. (6 pts) Answer each question below and **give a brief explanation**.

a) Is it possible for a nonhomogeneous system of seven equations in six unknowns to have a unique solution for some set of right-hand side of constants?

b) Is it possible for the system in part a) to have exactly 6 solutions for some set of right-hand side constants?

3. (4 pts) Let A be a 7×5 matrix.

a) If A has a 2-dimensional null space, what is the dimension of the row space of A ?

b) If A has a 2-dimensional null space, what is the dimension of the null space of A^T ?